

To design and verify a Differential Pair with PMOS Current Mirror Active Load in GF180MCU technology that satisfies gain, power, output common-mode voltage, and output swing specifications using the gm/ID design methodology.

Technology

GF180MCU PDK

TT Corner

Temperature = 27°C

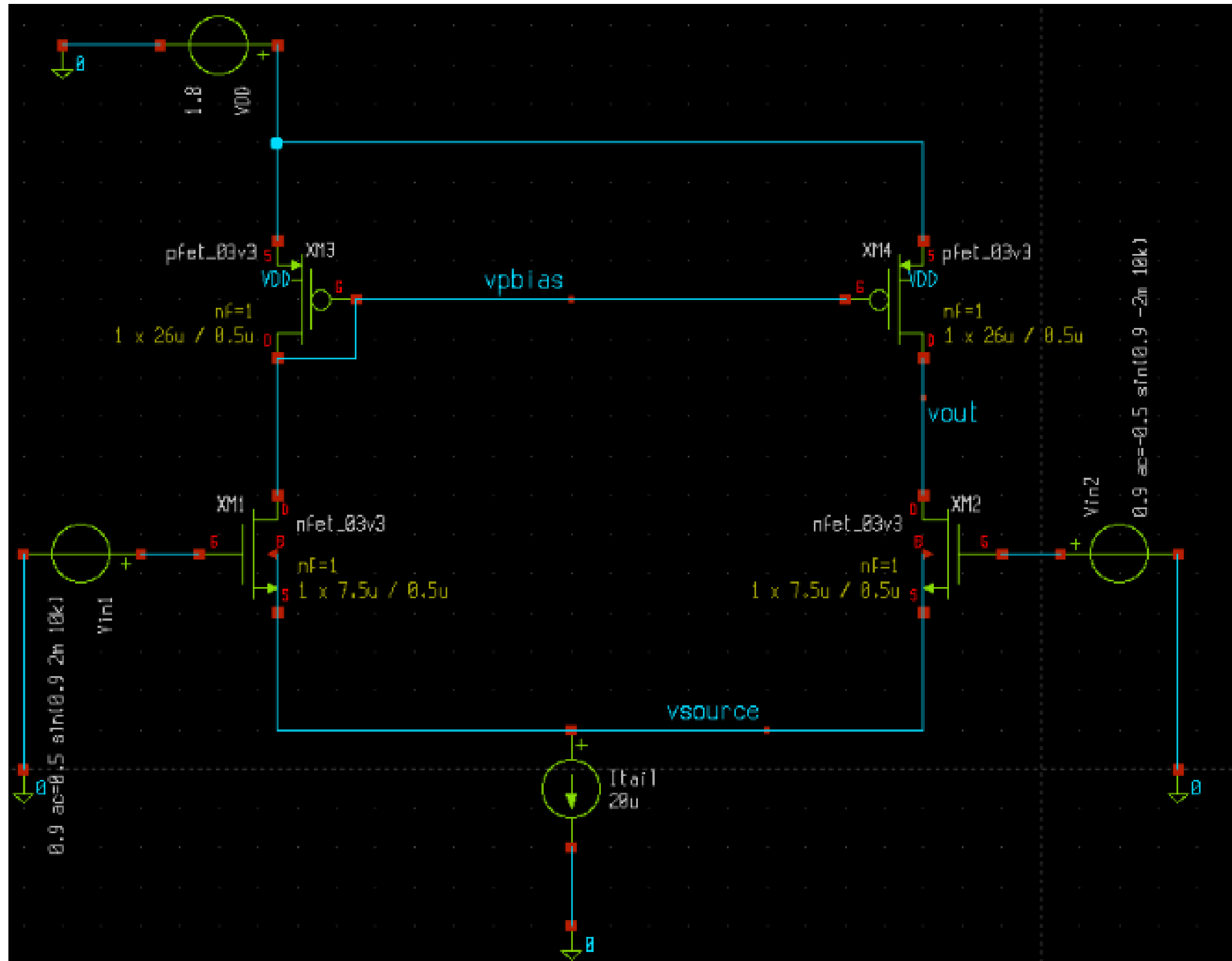
Supply

VDD = 1.8 V

Specifications

- Differential Gain $A_d \geq 50 \text{ V/V}$
- Tail Current $I_{tail} = 20 \text{ } \mu\text{A}$
- Power Consumption $\leq 40 \text{ } \mu\text{W}$
- Output Common Mode Voltage $0.8 \text{ V} - 1.2 \text{ V}$
- Single-Ended Output Swing $\geq \pm 300 \text{ mV}$
- All MOSFETs Operate in Saturation Region

Schematic

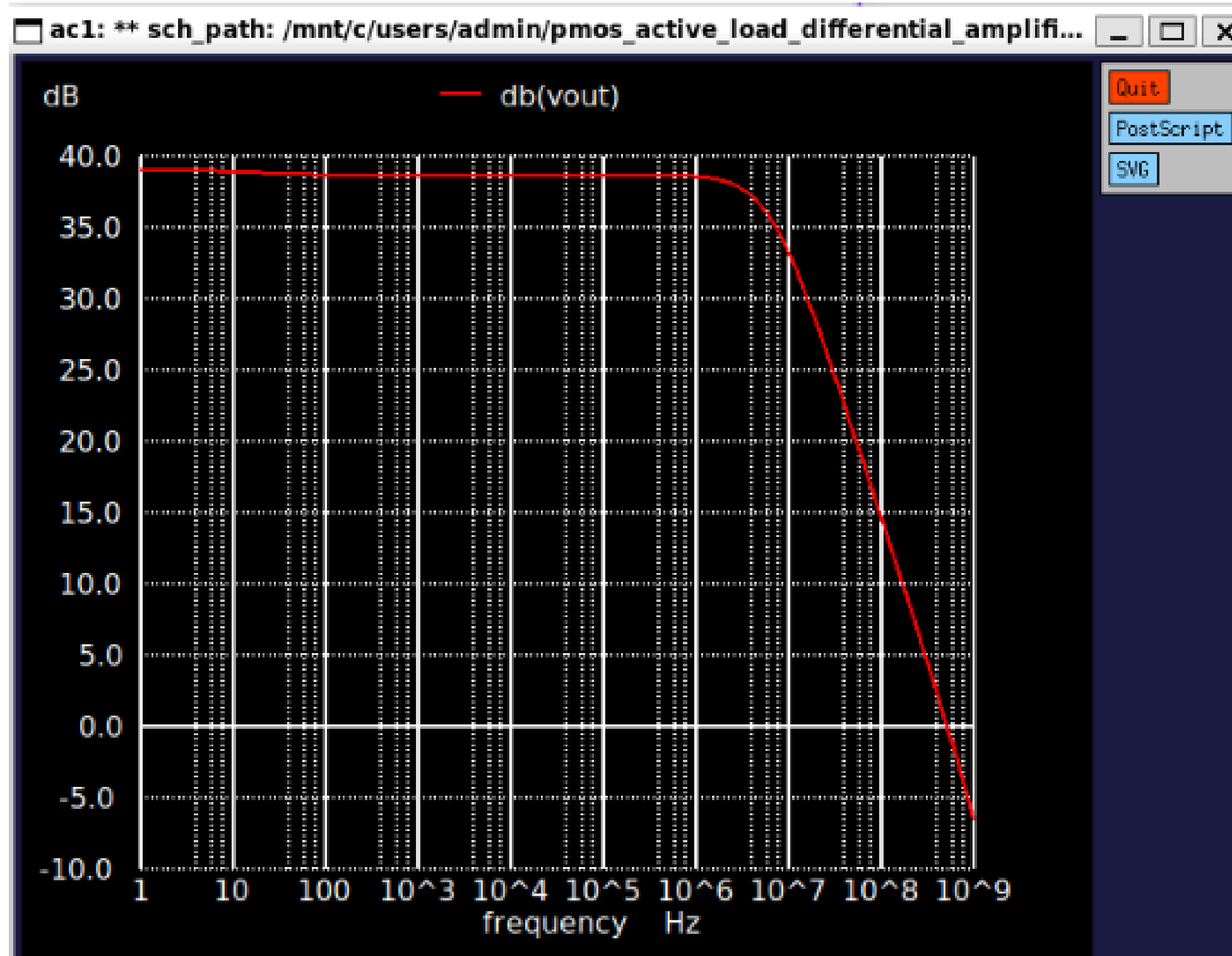


Operating Point Verification

```
pmos_active_load_differential_amplifier.spice" ... _ _ X
No. of Data Rows : 5008
ngspice 1 -> setplot
List of plots available:

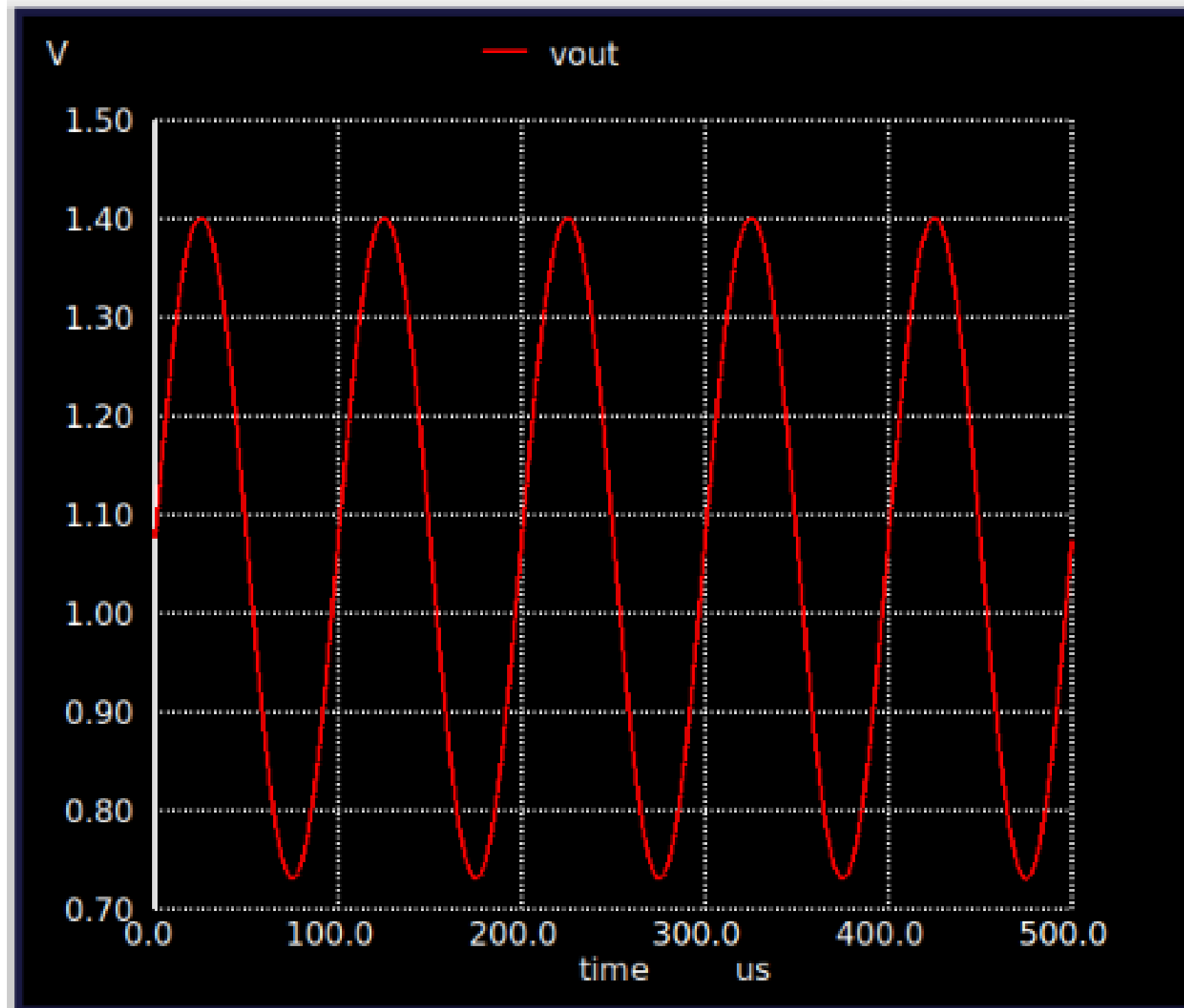
Current tran1 ** sch_path: /mnt/c/users/admin/pmos_active_load_differential_amplifier.sch (Transient Analysis)
op1 ** sch_path: /mnt/c/users/admin/pmos_active_load_differential_amplifier.sch (Operating Point)
ac1 ** sch_path: /mnt/c/users/admin/pmos_active_load_differential_amplifier.sch (AC Analysis)
const Constant values (constants)
ngspice 2 -> setplot op1
ngspice 3 -> print all
net1 = 1.800000e+00
net2 = 9.000000e-01
net3 = 9.000000e-01
vdd = 1.540033e+00
vdd#branch = -2.00000e-05
vin1#branch = 0.000000e+00
vin2#branch = 0.000000e+00
vout = 1.075367e+00
vpbias = 1.075367e+00
vsource = 1.504258e-01
ngspice 4 -> █
```

Gain Verification



Output swing Verification

tran1: ** sch_path: /mnt/c/users/admin/pmos_active_load_differential_ampli...



Specification Verification

- ✓ Differential Gain = 89 V/V (39 dB)
- ✓ Power Consumption = 36 μ W
- ✓ Tail Current = 20 μ A (Requirement: 20 μ A)
- ✓ Output Common-Mode Voltage (V_{OCM}) = 1.075 V
- ✓ Single-Ended Output Swing = $\pm$ 335 mV
- ✓ All MOSFETs Operate in Saturation
- ✓ Specifications Successfully Met